

# Perceived treatment group affects behavioral and neural responses to visceral pain in a deceptive placebo study

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## Abstract

**Background** To assess effects of perceived treatment (i.e. drug vs placebo) on behavioral and neural responses to rectal pain stimuli delivered in a deceptive placebo condition. **Methods** This fMRI study analyzed the behavioral and neural responses during expectation-mediated placebo analgesia in a rectal pain model. In  $N = 36$  healthy subjects, the blood oxygen level-dependent (BOLD) response during cued anticipation and painful stimulation was measured after participants were informed that they had a 50% chance of receiving either a potent analgesic drug or an inert substance (i.e., double-blind administration). In reality, all received placebo. We compared responses in subjects who retrospectively indicated that they received the drug and those who believed to have received placebo. **Key Results** 55.6% ( $N = 20$ ) of subjects believed that they had received a placebo, whereas 36.1% ( $N = 13$ ) believed that they had received a potent analgesic drug. Subjects who were uncertain (8.3%,  $N = 3$ ) were excluded. Rectal pain-induced discomfort was significantly lower in the perceived drug treatment group ( $P < 0.05$ ), along with significantly reduced activation of the insular, the posterior and anterior cingulate cortices during pain anticipation, and of the anterior cingulate cortex

during pain (all  $P < 0.05$  in regions-of-interest analyses). **Conclusions & Inferences** Perceived treatment constitutes an important aspect in placebo analgesia. A more refined understanding of individual treatment expectations and perceived treatment allocation has multiple implications for the design and interpretation of clinical trials and experimental studies on placebo and nocebo effects.

**Keywords** expectations, functional magnetic resonance imaging, placebo response, research design, visceral pain.

## INTRODUCTION

Whereas placebo effects in somatic pain are increasingly well understood,<sup>1,2</sup> comparatively little research has been conducted in the context of visceral or pelvic pain.<sup>3,4</sup> Given the high prevalence of chronic abdominal pain in various clinical conditions, including the functional gastrointestinal disorders, studies addressing placebo analgesia using clinically relevant models of visceral pain are highly relevant.<sup>4–6</sup> In fact, placebo interventions appear to be capable of producing clinically relevant benefits in irritable bowel syndrome (IBS).<sup>7,8</sup> At the same time, high placebo response rates in clinical trials have been felt to hamper progress in the identification of successful pharmacological treatment options for abdominal pain in IBS,<sup>1</sup> which further underscores the broad implications of a more refined understanding of placebo effects in visceral pain.

Knowledge about the pivotal role of expectations as a major psychological mechanism mediating placebo effects is rapidly accumulating. Indeed, the magnitude of placebo effects has been found to be proportional to the degree of certainty regarding the probability of

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receiving active drug in healthy subjects<sup>9,10</sup> and patients.<sup>11,12</sup> These experimental findings are consistent with evidence that placebo response rates in clinical trials are influenced by the number of active treatment arms in a trial,<sup>13,14</sup> presumably by changing patients' expectations regarding the probability of relief. Furthermore, analyses of trials in pediatric psychopharmacology<sup>15–17</sup> revealed that clinical evaluations were affected by perceived treatment allocation even when the child was in fact in the placebo group. At the same time, allocation concealment has been found to be an important problem.<sup>18</sup> For example, correct guesses regarding individual treatment assignment were significantly greater than chance<sup>15,16</sup> or differed significantly between patients in the treated and placebo groups,<sup>19</sup> which led to calls for more consistent analysis of effects of blinding and perceived treatment allocation in randomized controlled trials.<sup>18–20</sup>

Data from two placebo-controlled trials in patients with pain following dental surgery suggested that perceived treatment group assignment constituted a critical determinant of the degree of analgesia.<sup>21</sup> In experimental placebo analgesia models, it remains unknown if and to what extent subjectively perceived treatment allocation is associated with responses to placebo treatment. Therefore, we analyzed a subset of data from a complex placebo analgesia study carried out in healthy subjects.<sup>9</sup> In this functional magnetic resonance imaging (fMRI) study, participants' expectancies regarding a potent analgesic drug which in reality was a placebo were experimentally manipulated. Herein, we assessed whether self-reported, post hoc perceived treatment group (i.e. drug vs placebo) is associated with altered behavioral and neural responses to rectal pain when subjects were told that the probability of receiving active treatment was 50%. We compared behavioral and neural responses to painful rectal stimuli in subjects receiving placebo, but thought that they had received the drug ("perceived drug treatment group"), and those who correctly believed to have received placebo ("perceived placebo treatment group"), to test the hypothesis that perceived treatment with an analgesic drug is paralleled by reduced behavioral and neural responses to pain stimuli in a visceral placebo analgesia paradigm.

## MATERIALS AND METHODS

### Inclusion and exclusion criteria

Healthy men and women were recruited by local advertisement. General exclusion criteria included age <18 years and >45 years,

body mass index <18 or >27, any concurrent medical condition, including neurological, psychiatric, cardiovascular, immunological, and endocrine conditions, evidence of structural brain abnormality upon structural MRI scan; MRI-specific exclusion criteria (i.e., phobic anxiety, claustrophobia, ferromagnetic implantations). Only women on oral contraceptives were included to reduce potential confounding by menstrual cycle phase, and postmenopausal status was excluded by the age cut-off to reduce putative effects of female hormonal status and create a fairly homogenous sample. All participants were evaluated digitally for anal tissue damage (e.g., painful hemorrhoids) which may interfere with balloon placement. A history of psychological conditions (based on self-report) or presently increased scores on the Hospital Anxiety and Depression Scale (HADS)<sup>22</sup> were also exclusionary. Frequency and severity of gastrointestinal complaints suggestive of any functional or organic gastrointestinal condition were assessed with a standardized questionnaire. Right-handedness was ensured using a validated questionnaire.<sup>23</sup> Pregnancy was excluded by commercially available urinary test. The study protocol was approved by the local Ethics Committee (University Hospital of Essen, Germany). All participants gave written informed consent, were paid for their participation, and were at the conclusion of experiments debriefed about the deceptive aspects of the study.

### Study design

The study was comprised of two study days. Firstly, an intravenous indwelling catheter was affixed in the non-dominant arm, and rectal perceptual and pain thresholds were determined. Next, a structural MRI scan was completed. Finally, visceral pain-induced brain activation was measured with fMRI in a within-subject, repeated measures design with three expectation conditions (i.e., three separate scanning sessions, randomized order, separated by 30-min breaks) using deceptive instructions. In the condition analyzed herein, participants were informed that they had a 50% chance of receiving either an intravenous infusion containing a potent analgesic drug or an inert substance (sodium chloride) as administration was supposedly accomplished in a double-blind fashion. The two other conditions, reported elsewhere,<sup>9</sup> consisted of a 100% probability condition (i.e., instruction that a highly potent analgesic drug was administered) and a 0% probability condition (i.e., instruction that sodium chloride was administered). In reality, participants received sodium chloride in all three conditions. In all conditions, subjects received rectal distensions at individual discomfort pressures determined on the first study day. Immediately following each scanning session, subjects rated distensions using visual analogue scales (VAS), and following the 50% condition analyzed herein, perceived treatment arm was additionally assessed using a standardized questionnaire (see Methods section "Questionnaires"). Functional magnetic resonance imaging was used to measure BOLD response during pain anticipation and pain. A total of eight painful distensions (duration each 31 s) were accomplished in each condition. Pauses in-between distensions lasted 30 s. Distensions were preceded by a brief auditory cue ("warning signal") in form of a short beep delivered at pseudo-randomized intervals (3 s, 6 s, 9 s, 12 s) prior to initiation of balloon distension. At the end of each distension, a distinctly different short beep was implemented to inform subjects about the completion of the stimulation interval.

### Rectal distensions

Rectal distensions were carried out using a pressure-controlled barostat system (modified ISOBAR 3 device, G & J Electronics,

Ontario, Canada). Briefly, perception and pain thresholds were determined using staircase distensions with random pressure increments of 2–10 mmHg. Subjects were prompted to rate the sensation as follows: 1 = no perception, 2 = doubtful perception, 3 = sure perception, 4 = little discomfort, 5 = severe discomfort, still tolerable, and 6 = pain, not tolerable. The threshold for first perception was defined as the pressure when the rating changed from 2 to 3; the pain threshold as the pressure at which the rating changed from 5 to 6. For repeated distensions during the fMRI study, the highest distension pressure corresponding to ratings of 5 ('severe discomfort') was chosen. The maximal distension pressure was set at a pressure level of 50 mmHg.

## Questionnaires

Following delivery of painful stimuli, VAS were completed outside the scanner to quantify how painful subjects rated the distensions delivered during scanning, how much urge-to-defecate they felt during distensions, and how much distension-induced discomfort they experienced (VAS ends were defined as 0 = 'none' to 100 = 'very much'). In addition, perceived treatment arm was assessed by asking subjects to indicate on a standardized questionnaire whether they believed to have received drug or placebo or whether they were unsure about the treatment they had received (i.e., 'What do you believe you have received in the preceding session?'). Possible answers were: 'Inert substance/placebo'; 'pain killer'; 'not sure'.

For trait variables, measured on the first study day, symptoms of anxiety and depression were assessed with the HADS,<sup>22</sup> chronic stress with the Perceived Stress Questionnaire (PSQ),<sup>24</sup> personality traits with the NEO-Five Factor Inventory (NEO-FFI),<sup>25</sup> and emotional distress with the Symptom Checklist-90-Revised (SCL-90-R).<sup>26</sup>

For statistical analysis of questionnaire data, independent samples *t*-tests or Chi-squared tests (for dichotomous variables) were computed. In all analyses, the alpha level for significance was set at 0.05 and results are shown as mean  $\pm$  standard error of the mean (SEM).

## FMRI imaging and analyses

All MR images were acquired using a 1.5 T MR (Sonata, Siemens, Erlangen, Germany) with a standard head coil. A 3D FLASH sequence (TR 10 ms, TE 4.5 ms, flip angle 30°, FOV 240 mm, matrix 512, slice-thickness 1.5 mm) was acquired. Blood oxygen level-dependent (BOLD) contrast images were acquired using an echo-planar technique (TR 3100 ms, TE 50 ms, flip angle 90°, FOV 240 mm, and matrix 64) with 34 transversal slices angulated in direction of the corpus callosum with a thickness of 3 mm and a 0.3 mm slice gap. For analysis, SPM 05 software (Wellcome Department of Cognitive Neurology, London, UK) was used. Prior to statistical analysis, images were realigned to the mean, normalized to a standard EPI-template as implemented in SPM 05 and finally smoothed with an isotropic Gaussian kernel of 9 mm. Data were also subjected to high- and low-pass filtering and correction for temporal autocorrelations (based on a first-order autoregressive model).

Data analysis was performed using a general linear model approach. All regressors were obtained by convolving a box-car function of the event duration with the canonical hemodynamic response function implemented in SPM. Specific effects were tested with appropriate linear contrasts of the parameter estimates for the different regressors resulting in a *t*-statistic for each voxel. After model estimation, the ensuing first-level contrast images from each subject were used for second-level analysis

treating individual subjects as a random factor and including non-sphericity correction. Data from the 50% probability condition were analyzed in two steps. (i) Activation in ROIs was analyzed in the whole sample of  $N = 36$  healthy subjects using one-sample *t*-tests to describe cued anticipation and pain-related neural activation irrespective of perceived treatment group. (ii) Subsequently, the neural response in the two groups who differed in perceived treatment, i.e., 'perceived drug treatment group' vs 'perceived placebo treatment group' was compared using two-sample *t*-tests with distension pressure as a covariate of no interest (to control for inter-individual differences in pain thresholds and accordingly distension pressures). Subjects who were unsure regarding their treatment ( $N = 3$ ) were excluded from this analysis.

Based on previous placebo analgesia and visceral pain studies, ROIs included the thalamus, insula, somatosensory cortex, cingulate cortex, and prefrontal cortices [dorsolateral prefrontal cortex (DLPFC); ventrolateral prefrontal cortex (VLPFC); orbito-frontal cortex (OFC)]. We used small volume correction (SVC) with familywise error (FWE) correction for multiple comparisons at a level of  $P < 0.05$ . Small volume correction was performed with templates constructed from the automated anatomical labeling (aal) toolbox in SPM.<sup>27</sup> All results are given as MNI coordinates.

## RESULTS

### Participants

Thirty-six subjects (18 females; 18 males) participated. All participants (100%) had a high school degree, 58.3% ( $N = 21$ ) were currently university students and the rest were employed full-time (22.2%,  $N = 8$ ) or part-time (19.4%,  $N = 7$ ). Twenty-five percent ( $N = 9$ ) were smokers. Self-perceived health was good (27.8%,  $N = 10$ ) or very good (72.2%,  $N = 26$ ). Approximately half of the sample (44.4%,  $N = 16$ ) was in a steady relationship or married, the remainder of subjects were either single (50%,  $N = 18$ ) or divorced (5.6%,  $N = 2$ ). Mean depression and anxiety scores were well within the normal range (HADS depression score:  $1.08 \pm 0.21$ ; anxiety score:  $3.28 \pm 0.36$ ). Mean rectal sensory threshold was  $16.1 \pm 0.4$  mmHg and mean rectal pain threshold was  $39.0 \pm 1.6$  mmHg.

### VAS ratings and rectal pain-induced neural activation in all subjects

Subjective evaluations of rectal distensions, assessed with VAS, were  $52 \pm 5$  mm for pain,  $70 \pm 3$  mm for discomfort, and  $66 \pm 4$  mmHg for urge-to-defecate, indicating that repeated distensions in the scanner were perceived as moderately aversive on average.

To initially describe rectal distension-induced neural activation, we performed one-sample *t*-tests on data from the whole sample of  $N = 36$  subjects. We observed activation of bilateral insular cortices, somatosensory

cortices (bilateral postcentral gyri and inferior parietal lobules), left mid-cingulate cortex, and inferior frontal cortex (VLPFC, DLPFC) (Table 1). No significant activations within the ROIs were observed during pain anticipation. Results of supplementary whole-brain analyses at  $P < 0.001$  are given in the legend to Table 1.

### Perceived treatment allocation following blinded placebo administration

Following delivery of painful stimuli and VAS ratings, perceived treatment arm was assessed. Of the total  $N = 36$  subjects,  $N = 20$  (55.6%) believed that they had received a placebo,  $N = 13$  (36.1%) felt that they had received a potent analgesic drug, and  $N = 3$  (8.3%) were unsure what they had received. Subjects who were unsure regarding treatment were excluded from further analyses, and all subsequent analyses compared the "perceived drug treatment group" and the "perceived placebo treatment group". There were no differences between these groups in any sociodemographic or psychological characteristics, including chronic stress, psychological disturbances, and personality factors (Table 2). Interestingly, mean rectal pain threshold, assessed on the first study day, was significantly lower in the perceived drug treatment group ( $P < 0.05$ , Table 2).

As a supplementary analysis, we used a median split to divide subjects into those who reported high vs low discomfort to assess whether the actual experience of pain could play a role in perceived post hoc group assignment. In subjects with low discomfort, the number of individuals who believed to have received

the drug (7 of 15) was similar to those who believed to have received a placebo (8 of 15). In subjects with high discomfort, fewer subjects believed to have received the drug (6 of 18) compared with those who believed to be in the placebo group (12 of 18). However, a Chi-squared test on this distribution was not significant ( $P = 0.438$ ).

### Behavioral and neural responses to blinded placebo administration in the two perceived treatment groups

In the perceived drug treatment group, rectal pain-induced discomfort ratings were significantly lower when compared with the perceived placebo treatment group ( $P < 0.05$ ), whereas pain and urge-to-defecate ratings did not differ significantly (Fig. 1).

Comparing neural activation in specific ROIs using two-sample  $t$ -tests with pain threshold as a covariate of no interest, the perceived drug treatment group revealed significantly reduced activation of the right insula ( $x = 30$ ,  $y = -24$ ,  $z = 22$ ,  $t = 4.99$ ,  $p_{FWE} < 0.05$ , Fig. 2A), the posterior cingulate cortex ( $x = 10$ ,  $y = -40$ ,  $z = 28$ ,  $t = 4.63$ ,  $p_{FWE} < 0.05$ , Fig. 2B), and the anterior cingulate cortex ( $x = 10$ ,  $y = 34$ ,  $z = 12$ ,  $t = 4.15$ ,  $p_{FWE} < 0.05$ , Fig. 2C) when compared with the perceived placebo treatment group during cued pain anticipation. During pain, activation of the anterior cingulate cortex [activation maximum at the border of subgenual (sACC) and pregenual (pACC) anterior cingulate cortex (ACC)]<sup>28</sup> was significantly reduced in the perceived drug treatment group ( $x = 12$ ,  $y = 30$ ,  $z = -12$ ,  $t = 4.41$ ,  $p_{FWE} < 0.05$ , Fig. 2D). Supplementary

**Table 1** Rectal pain-induced neural activation of ROIs over all  $N = 36$  subjects

| Brain region of interest (ROI)   | Coordinates |     |     |     | $t$ -value |
|----------------------------------|-------------|-----|-----|-----|------------|
|                                  | $H$         | $x$ | $y$ | $z$ |            |
| Pain                             |             |     |     |     |            |
| Insula                           | R           | 34  | 22  | 4   | 10.45      |
|                                  | L           | -38 | 18  | -6  | 8.03       |
| Prefrontal cortex (VLPFC, DLPFC) | R           | 48  | 46  | 0   | 6.61       |
|                                  | R           | 42  | 50  | 16  | 5.12       |
| Posterior lobe (SI, SII)         | R           | 52  | -30 | 28  | 7.15       |
|                                  | L           | -66 | -20 | 22  | 5.29       |
| Inferior parietal lobule, SII    | R           | 58  | -36 | 54  | 4.97       |
|                                  | L           | -58 | -42 | 48  | 4.97       |
| Mid-cingulate cortex             | L           | 6   | -24 | 28  | 5.66       |
|                                  | L           | 8   | 4   | 48  | 4.88       |

One-sample test assessing cued rectal pain-induced neural activation in ROIs, computed on data from all  $N = 36$  healthy subjects using one-sample  $t$ -test. All  $P < 0.05$  based on ROI analysis using SVC with FWE correction (no significant activations were found for cued anticipation at  $p_{FWE} < 0.05$ ). Additional one-sample  $t$ -test results of whole-brain statistics at  $P < 0.001$  uncorrected during pain anticipation: right insula ( $x = 36$ ,  $y = 22$ ,  $z = 6$ ,  $t = 3.98$ ); during pain right thalamus ( $x = 10$ ,  $y = -12$ ,  $z = -2$ ); left posterior mid-cingulate cortex ( $x = 10$ ,  $y = 4$ ,  $z = 48$ ,  $t = 4.01$ ); left medial frontal gyrus ( $x = 6$ ,  $y = 24$ ,  $z = 52$ ,  $t = 4.19$ ); bilateral cerebellum ( $x = 14$ ,  $y = -72$ ,  $z = -48$ ,  $t = 5.20$ ;  $x = -24$ ,  $y = -66$ ,  $z = -28$ ,  $t = 5.24$ ). H, Hemisphere with activation; R, right asymmetrical activation; L, left asymmetrical activation; DLPFC, dorsolateral prefrontal cortex; VLPFC, ventrolateral prefrontal cortex.



**Table 2** Sociodemographic and psychological group characteristics

|                                          | Perceived drug treatment group | Perceived placebo treatment group | <i>P</i> |
|------------------------------------------|--------------------------------|-----------------------------------|----------|
| Females (N) : males (N)                  | 7 : 6                          | 11 : 9                            | ns       |
| Age, years                               | 25.23 ± 2.03                   | 27.10 ± 1.71                      | ns       |
| Body mass index                          | 23.38 ± 1.09                   | 23.60 ± 0.73                      | ns       |
| HADS depression score                    | 1.54 ± 0.42                    | 0.75 ± 0.19                       | ns       |
| HADS anxiety score                       | 2.85 ± 0.59                    | 3.65 ± 0.39                       | ns       |
| SCL-90-R global severity index           | 48.83 ± 1.79                   | 43.89 ± 1.77                      | ns       |
| SCL-90-R positive symptom distress index | 48.33 ± 2.10                   | 44.26 ± 1.89                      | ns       |
| SCL-90-R positive symptom total          | 49.00 ± 1.51                   | 44.84 ± 1.85                      | ns       |
| NEO-FFI neuroticism                      | 2.42 ± 0.18                    | 2.44 ± 0.13                       | ns       |
| NEO-FFI extraversion                     | 4.06 ± 0.30                    | 4.58 ± 0.11                       | ns       |
| NEO-FFI openness                         | 4.54 ± 0.20                    | 4.35 ± 0.12                       | ns       |
| NEO-FFI agreeableness                    | 4.61 ± 0.14                    | 4.60 ± 0.13                       | ns       |
| NEO-FFI conscientiousness                | 4.32 ± 0.30                    | 4.34 ± 0.20                       | ns       |
| PSQ tension                              | 8.08 ± 0.54                    | 8.21 ± 0.39                       | ns       |
| PSQ demands                              | 9.83 ± 0.81                    | 9.26 ± 0.49                       | ns       |
| PSQ joy                                  | 8.92 ± 0.75                    | 7.63 ± 0.47                       | ns       |
| PSQ worries                              | 7.08 ± 0.63                    | 6.95 ± 0.35                       | ns       |
| Rectal sensory threshold, mmHg           | 15.15 ± 1.10                   | 15.70 ± 0.81                      | ns       |
| Rectal pain threshold, mmHg              | 34.92 ± 1.64                   | 40.80 ± 1.45                      | 0.013    |

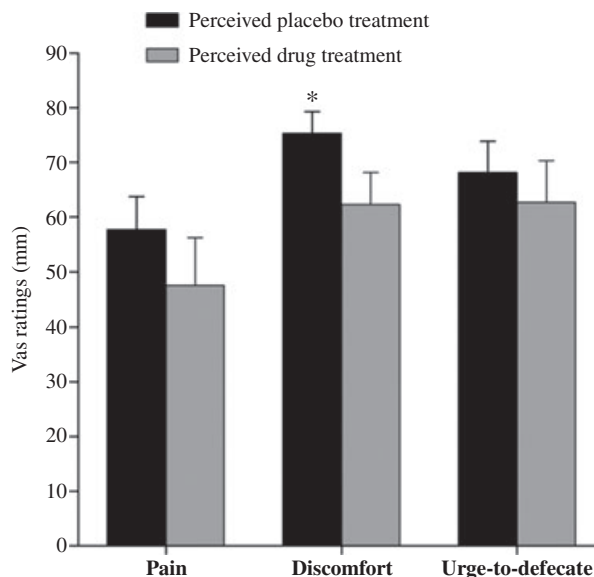
Psychosocial characteristics of the two groups who differed in perceived treatment, i.e., the “perceived drug treatment group” (*N* = 13) vs the “perceived placebo treatment group” (*N* = 20) were compared with independent two-sample *t*-tests or Chi-squared Test (gender distribution). Subjects who were unsure regarding their treatment (*N* = 3) were excluded from analysis. ns = not significant.

All data are shown as mean ± SEM. German versions of validated questionnaires were utilized: Anxiety and depression symptoms with the Hospital Anxiety and Depression Scale (HADS); emotional distress with the symptom checklist revised (SCL-90-R); personality traits with the NEO-Five Factor Inventory (NEO-FFI); chronic perceived stress with the perceived stress questionnaire (PSQ), for references, see Method section.

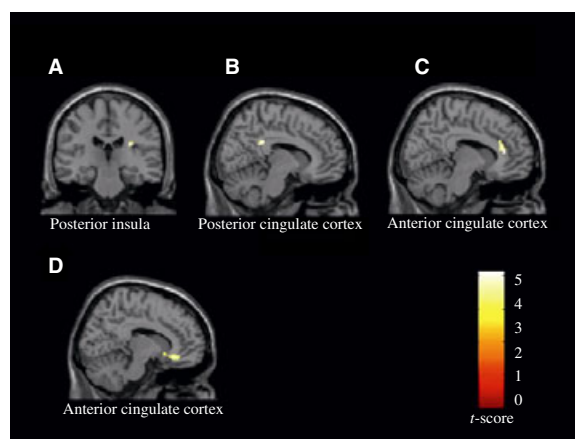
whole-brain analyses (*P* < 0.001 uncorrected) demonstrated reduced activation in multiple additional brain areas relevant for pain processing in the perceived drug treatment group compared with the perceived placebo treatment group (Table 3).

## DISCUSSION

A more refined understanding of the role of individual treatment expectations and perceived treatment allocation has implications for the design and interpretation of clinical trials and experimental studies on placebo and nocebo effects.<sup>29–31</sup> However, it remains unknown if post hoc perceived treatment allocation is associated with altered behavioral and neural responses to placebo treatment in the context of experimental visceral pain. In this study on visceral placebo analgesia, subjects experiencing painful rectal distension were given instructions that the probability of receiving



**Figure 1** Visual analogue scale ratings of rectal distension-induced pain, discomfort, and urge-to-defecate. In the perceived drug treatment group, rectal pain-induced discomfort ratings were significantly lower when compared with the perceived placebo treatment group (*P* < 0.05, independent two-sample *t*-test), whereas pain and urge-to-defecate ratings did not differ significantly. Data are shown as mean ± SEM.



**Figure 2** Comparison of cortical activation during cued-anticipation (A, B, C) and rectal pain (D) in the perceived drug treatment and the perceived placebo treatment group. Two-sample *t*-tests revealed significantly reduced activation of (A) the right insula, (B) the posterior cingulate cortex, and (C) the anterior cingulate cortex in the perceived drug treatment group compared with the perceived placebo treatment group during cued pain anticipation. During pain, activation of the anterior cingulate cortex was significantly reduced in the perceived drug treatment group (D). All results of regions-of-interest analysis are using two-sample *t*-tests with small-volume correction with family-wise error correction (FWE; *P* < 0.05).

ing a potent pain killer was 50%, when in reality a placebo was always administered. Retrospective ratings regarding perceived treatment allocation revealed that about one-third of participants (36.1%) believed to

**Table 3** Whole-brain analysis of group differences in rectal pain-induced neural activation in the perceived drug treatment < the perceived placebo treatment group

| Brain regions                          | Coordinates |     |     |     | t-value |
|----------------------------------------|-------------|-----|-----|-----|---------|
|                                        | H           | x   | y   | z   |         |
| Anticipation                           |             |     |     |     |         |
| Posterior insula                       | R           | 30  | -24 | 22  | 4.99*   |
| Posterior cingulate cortex             | R           | 10  | -40 | 28  | 4.63*   |
| Anterior cingulate cortex              | R           | 10  | 34  | 12  | 4.15*   |
| Superior frontal gyrus                 | L           | -14 | 56  | 22  | 4.24    |
| Thalamus                               | L           | -14 | -6  | 12  | 4.15    |
| Superior temporal gyrus                | L           | -52 | -46 | 16  | 4.83    |
| Middle temporal gyrus                  | R           | 52  | -16 | -18 | 4.37    |
| Middle temporal gyrus                  | L           | -32 | -12 | -14 | 4.00    |
| Supplementary motor area               | L           | -6  | 10  | 70  | 4.09    |
| Putamen                                | R           | 18  | 18  | -6  | 4.39    |
| Cerebellum                             | R           | 46  | -60 | -24 | 4.38    |
| Pain                                   |             |     |     |     |         |
| Caudate nucleus                        | R           | 18  | 20  | -8  | 4.68    |
| Anterior cingulate cortex              | R           | 12  | 30  | -12 | 4.41*   |
| Inferior frontal gyrus/anterior insula | L           | -26 | 34  | 0   | 4.06    |

Two-sample test results of whole-brain analysis ( $P < 0.001$  uncorrected, with distension pressure as a covariate of no interest) assessing cued rectal pain-induced neural activation in the perceived drug treatment group < perceived placebo treatment group. No significant activations were found for the reverse contrast (perceived drug treatment > perceived placebo treatment). \*Results that reached significance in ROI analyses with small-volume correction using familywise error correction (FWE) (all  $p_{\text{FWE}} < 0.05$ ) and are shown in Fig. 2. H, Hemisphere with activation; R, right asymmetrical activation; L, left asymmetrical activation.

have received active drug ("perceived drug treatment group"), whereas approximately half of the sample (55.6%) believed that they had received a placebo ("perceived placebo treatment group"). These groups differed both at the behavioral and neural levels in their responses to painful rectal stimuli. Subjects in the perceived drug treatment group demonstrated lower rectal distension-induced discomfort along with reduced activation of the insula and cingulate regions during cued anticipation and delivery of rectal pain stimuli.

The fact that these activation differences occurred in brain areas known to be key components of the "central pain matrix", relevant for the integration of sensory and affective aspects of the pain response, supports that retrospective assessment of perceived treatment group and stimulus evaluation were in fact preceded by altered processing of ascending nociceptive input and were not simply a reporting bias. These activation differences were also clearly not attributable to inter-individual differences in pain rectal pain thresholds (or accordingly, actual distension pressures) as these were statistically taken into account in group comparisons. The cingulate and insular cortices are part of a network responsible for the coordination of affective reactions to painful stimuli by encoding emotional, motivational, and cognitive demands.<sup>32,33</sup> The posterior cingulate cortex (PCC) is involved in orienting the body in response to sensory stimuli, including but not limited to nociceptive stimuli.<sup>28</sup> We

previously reported that rectal distension-induced activation, specifically of the PCC was affected by psychological stress in healthy subjects.<sup>34</sup> Furthermore, patients with IBS and a history of abuse, reported more pain and demonstrated greater MCC/PCC activation during rectal distensions.<sup>35</sup> The crucial relevance of ACC regions in visceral pain both in healthy subjects and patients with IBS has been demonstrated in numerous studies.<sup>36</sup> Interestingly, we previously reported that pregenual ACC activation correlated with anxiety symptoms in IBS patients.<sup>37</sup> Together, these previous findings support that reduced activation of the PCC and pregenual/subgenual ACC regions and of the insula in the group of placebo-treated subjects, who believed to be in the active drug group, may reflect reduced amplification of pain by emotions of fear and anxiety. This is consistent with our finding of reduced pain-induced discomfort, but unaltered pain intensity and urge-to-defecate ratings in the perceived drug treatment group. Discomfort is arguably a dimension of the subjective pain experience that is most strongly affected by affective and motivational aspects, relative to sensory-discriminatory components.

As we did not assess (pre)treatment expectations, we cannot determine whether retrospectively assessed perceived treatment allocation resulted from pre-existing differences in expectations (i.e., *prior* to the experience of painful stimuli) or arose *during* or *after* the experience of painful stimuli. Indeed, perceived treat-

ment assignment may at least in part reflect the perceived amount of symptom reduction, independently of the expectations of pain relief. This notion is supported by our observation that the percentage of subjects who believed to have received drug was lower in subjects who experienced more discomfort. This difference in distribution was not significant, but our sample size was clearly too small for meaningful subgroup analyses.

Nevertheless, in the context of an acupuncture trial, positive expectancy reportedly increased over the course of a trial,<sup>38</sup> which would support that the assessment of perceived treatment at the end of the treatment may constitute a valid criterion. Furthermore, our data are consistent with findings from two acupuncture trials where subjects who believed they received real acupuncture reported significantly less pain than those who thought they were in the placebo group.<sup>21</sup> The critical relevance of positive (and to a lesser extent negative) treatment expectations has been demonstrated both in clinical trials<sup>39</sup> and experimentally in healthy subjects,<sup>9,40,41</sup> and patients.<sup>42–44</sup> The present data complement and extend these findings by demonstrating for the first time that retrospectively assessed perceived treatment allocation constitutes an important aspect in placebo analgesia at both the behavioral and neural levels. In fact, future studies may even consider the possibility that subjectively perceived treatment arm or group could constitute an additional or alternative criterion to define “placebo responders” which is usually determined using a clinical response criterion (e.g., extent of pain reduction). Clearly, allocation concealment can be problematic in some randomized controlled trials,<sup>18</sup> and it will be important in both clinical and experimental studies involving placebo conditions or placebo study arms to consistently incorporate measures of both treatment expectations and perceived treatment allocation.

Given high inter-individual variability in placebo responses,<sup>45–47</sup> interest in putative variables defining “placebo responders” and/or predictor(s) of the placebo response is growing. In the present analysis, we found that mean rectal pain threshold, assessed prior to the placebo study, was significantly lower in the perceived drug treatment group. Based on this finding, one is tempted to speculate that increased pain sensitivity may be associated with a greater tendency and/or susceptibility to placebo modulation. However, given the present post hoc analysis, no cause–effect relationships can be established. Nevertheless, if future studies could replicate this finding in patients characterized by

increased visceral pain stimuli, such as IBS, such a connection would be of considerable scientific and clinical interest. Beyond a group difference in rectal pain threshold, we could not find any differences between the groups in the distribution of males and females and in various psychological state and trait variables. These negative findings are in accordance with the larger placebo literature which thus far has provided few reliable predictors in psychosocial measures.<sup>48–50</sup> On the other hand, there likely exist inter-individual variations in brain neurochemistry, e.g., in the endogenous opioidergic and/or dopaminergic systems, relevant to placebo and nocebo effects that we did not assess in our study.<sup>47,51,52</sup> Furthermore, a recent reanalysis of fMRI data suggested that the pattern of activation in specific brain networks during the anticipation and experience of pain are indeed predictors of individual differences in placebo responses.<sup>53</sup> Our analysis of fMRI data with regard to differences between groups confirms that perceived treatment with drug vs placebo is paralleled by differences in brain activation in brain regions that are part of the central pain matrix and could constitute an important predictor of placebo and/or drug effects in clinical trials.

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## DISCLOSURE

All authors declare no conflict of interest.

## AUTHOR CONTRIBUTIONS

VK analysis and interpretation of data; drafting of the article; SB acquisition of data; interpretation of data; critical revision of the article; UB conception and design; analysis and interpretation of data; critical revision of the article; MF acquisition of funding, acquisition of data, critical revision of the article; MS conception and design, interpretation of data, critical revision of the article; ERG acquisition of funding, conception and design, acquisition of data, critical revision of the article; SE acquisition of funding, conception and design, analysis and interpretation of data, critical revision of the article; All authors have given final approval of the final version submitted for peer review.

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